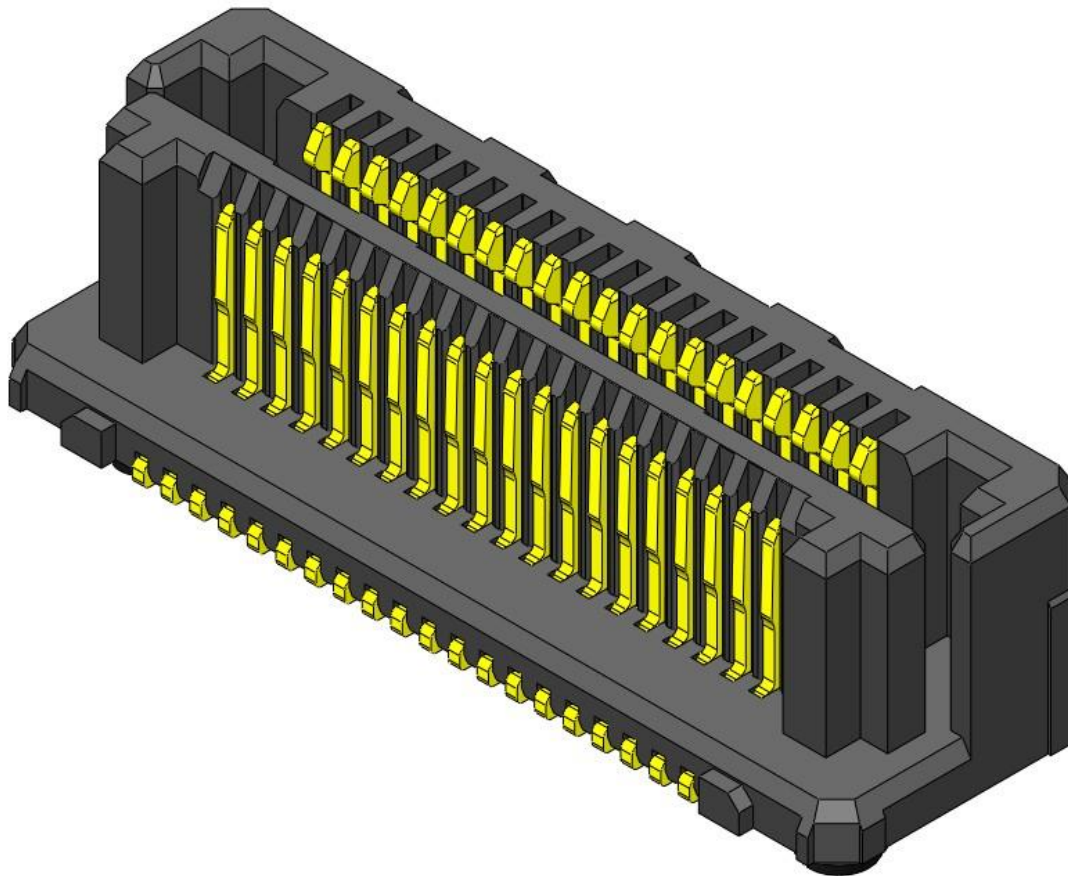


Series: LSHM 0,50mm (.0197") Razor Beam™ High Speed Hermaphroditic Terminal/Socket Strip

LSHM Series – Terminal/Socket, No Shield, Vertical Orientation



Other configurations available for:

Right Angle Application
Reverse Right Angle Application
With Shield Application

See www.samtec.com for more information.

Series: LSHM 0,50mm (.0197") Razor Beam™ High Speed Hermaphroditic Terminal/Socket Strip

1.0 SCOPE

1.1 This specification covers performance, testing and quality requirements for Samtec 0,50mm (.0197") Razor Beam™ High Speed Hermaphroditic Terminal/Socket Strip connectors. All information contained in this specification is for two LSHM Series connectors in a 2,50 mm (.098") mated height vertical configuration unless otherwise noted.

2.0 DETAILED INFORMATION

2.1 Product prints, footprints, catalog pages, test reports and other specific, detailed information can be found at www.samtec.com?LSHM.

3.0 TESTING

3.1 Current Rating: 2A (One Pins Powered Per Row)

3.2 Voltage Rating: 160 VAC

3.3 Operating Temperature Range: -55°C to +125°C

3.4 Electrical:

ITEM	TEST CONDITION	REQUIREMENT
Withstanding Voltage	EIA-364-20 (No Flashover, Sparkover, or Breakdown)	470 VAC
Insulation Resistance	EIA-364-21 (1000 MΩ minimum)	1,000 MΩ
Contact Resistance (LLCR)	EIA-364-23	Δ 15 mΩ maximum (Samtec defined)/ No damage

3.5 Mechanical:

ITEM	TEST CONDITION	REQUIREMENT	STATUS
Durability	EIA-364-09C	100 cycles	Pass
Random Vibration	EIA-364-28 Condition V, Letter B 7.56 G 'RMS', 50 to 2000 Hz, 2 hours per axis, 3 axis total, PSD 0.04	Visual Inspection: No Damage LLCR: Δ 15 mΩ maximum Event Detection: No interruption > 50 Nanoseconds	Pass
Mechanical Shock	EIA-364-27 100 G, 6 milliseconds, sawtooth wave, 11.3 fps, 3 shocks/direction, 3 axis (18 total shocks)	Visual Inspection: No Damage LLCR: Δ 15 mΩ maximum Event Detection: No interruption > 50 Nanoseconds	Pass
Normal Force	EIA-364-04	30 grams minimum for gold interface	Pass

Series: LSHM 0,50mm (.0197") Razor Beam™ High Speed Hermaphroditic Terminal/Socket Strip

3.6 Environmental:

ITEM	TEST CONDITION	REQUIREMENT	STATUS
Thermal Shock	EIA-364-32 Thermal Cycles: 100 (30 minute dwell) Hot Temp: 85°C Cold Temp: -55°C Hot/Cold Transition: Immediate	Visual Inspection: No Damage LLCR: Δ 15 m Ω DWV: 470 VAC IR: >1,000 M Ω	Pass
Thermal Aging (Temp Life)	EIA-364-17 Test Condition 4 @ 105°C Condition B for 250 hours	Visual Inspection: No Damage LLCR: Δ 15 m Ω DWV: 470 VAC IR: >1,000 M Ω	Pass
Cyclic Humidity	EIA-364-31 Test Temp: 25°C to 65°C Relative Humidity: 90 to 95% Test Duration: 240 hours	Visual Inspection: No Damage LLCR: Δ 15 m Ω DWV: 470 VAC IR: >1,000 M Ω	Pass
Gas Tight	EIA-364-36 Gas Exposure: Nitric Acid Vapor Duration: 60 min. Drying Temp.: 50°C +/- 3°C Measurements: Within 1 hour of Exposure	LLCR: Δ 15 m Ω	Pass

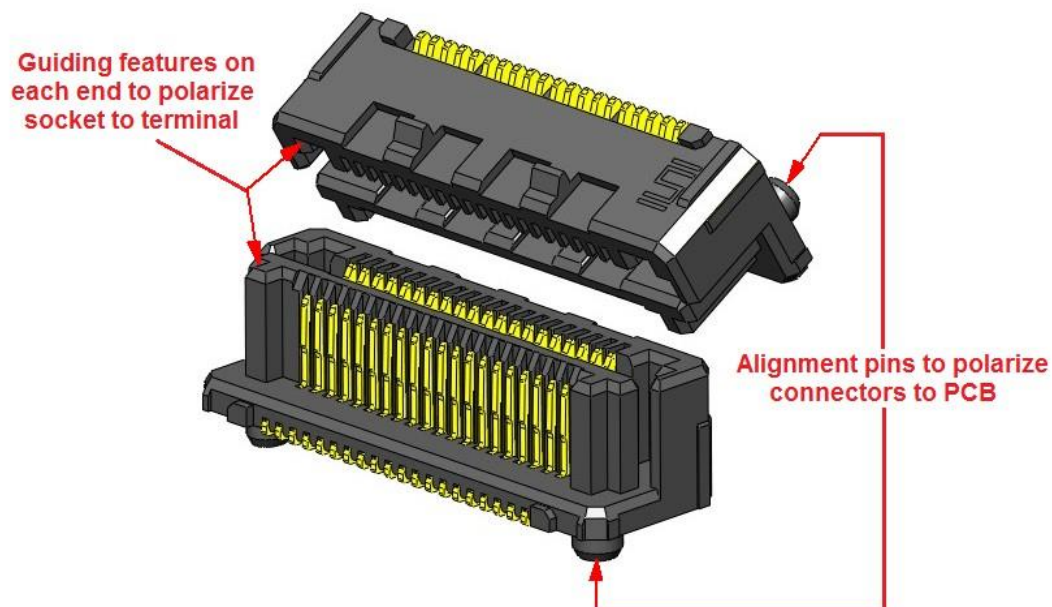
Series: LSHM 0,50mm (.0197") Razor Beam™ High Speed Hermaphroditic Terminal/Socket Strip

4.0 MATED SYSTEM

4.1 Mated views - Mated view information can be found at link below:

<http://suddendocs.samtec.com/prints/lshm%20mated%20document.pdf>

5.0 POLARIZING FEATURES



6.0 HIGH SPEED PERFORMANCE

6.1 Based on a 3 dB insertion loss

6.2 System Impedance: 50 ohm for single-ended and 100 ohm for differential pair

Stack Height	Single-Ended Signaling	Differential Pair Signaling
5 mm	11.5 GHz/23.0 Gbps	7.0 GHz/14.0 Gbps

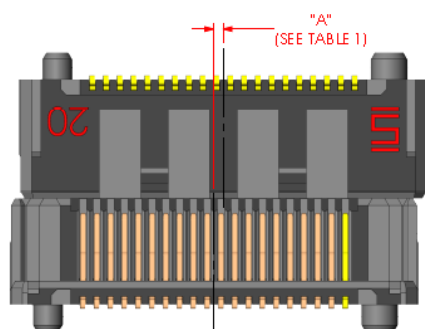
Series: LSHM 0,50mm (.0197") Razor Beam™ High Speed Hermaphroditic Terminal/Socket Strip

7.0 PROCESSING RECOMMENDATIONS

7.1 Mating Alignment Requirements:

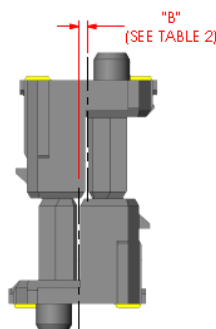
7.1.1 Allowable initial linear misalignment

TABLE 1	
ORIENTATION	"A"
DV TO DV	0.34 [.013]
RA TO RA	0.34 [.013]



INITIAL X AXIS LINEAR MISALIGNMENT

TABLE 2	
ORIENTATION	"B"
DV TO DV	0.20 [.008]
RA TO RA	0.40 [.016]



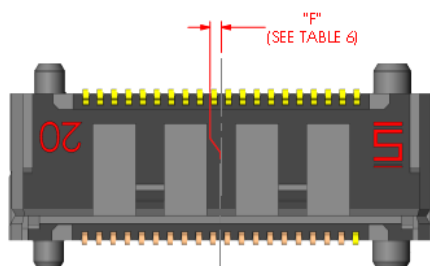
INITIAL Y AXIS LINEAR MISALIGNMENT

NON APPLICABLE

INITIAL Z AXIS LINEAR MISALIGNMENT

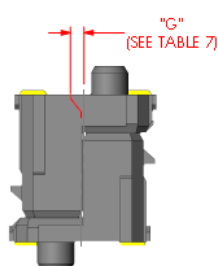
7.1.2 Allowable final linear misalignment

TABLE 6	
ORIENTATION	"F"
DV TO DV	0.05 [.002]
RA TO RA	0.05 [.002]



FINAL X AXIS LINEAR MISALIGNMENT

TABLE 7	
ORIENTATION	"G"
DV TO DV	0.05 [.002]
RA TO RA	0.10 [.004]



FINAL Y AXIS LINEAR MISALIGNMENT

SEE MATED VIEW

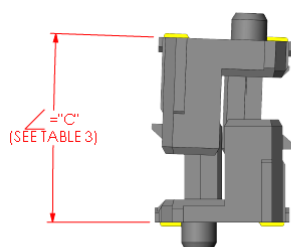
FINAL Z AXIS LINEAR MISALIGNMENT

Series: LSHM 0,50mm (.0197") Razor Beam™ High Speed Hermaphroditic Terminal/Socket Strip

7.2 Mating Angle Requirements

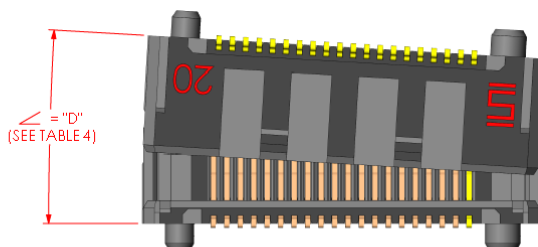
7.2.1 Allowable initial angular misalignment

TABLE 3	
ORIENTATION	"C" = DEG
DV TO DV	2.0
RA TO RA	4.0



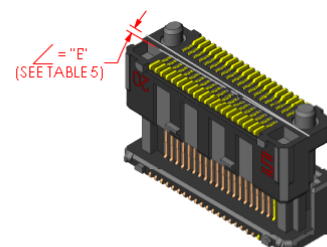
INITIAL X AXIS ANGULAR MISALIGNMENT

TABLE 4			
POSITION	"D" = DEG	POSITION	"D" = DEG
5	8.0	30	1.5
10	5.0	40	1.2
20	2.5	50	1.0



INITIAL Y AXIS ANGULAR MISALIGNMENT

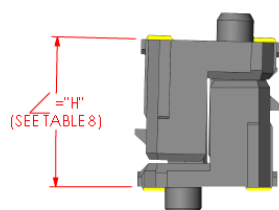
TABLE 5			
POSITION	"E" = DEG	POSITION	"E" = DEG
5	6.0	30	1.0
10	3.0	40	0.7
20	1.5	50	0.5



INITIAL Z AXIS ANGULAR MISALIGNMENT

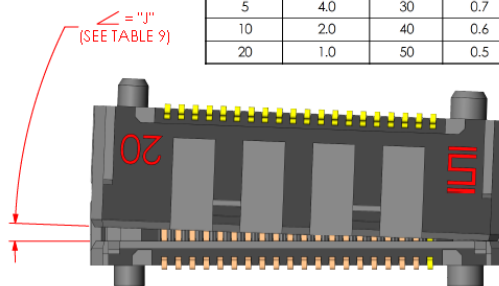
7.2.2 Allowable final angular misalignment

TABLE 8	
ORIENTATION	"H" = DEG
DV TO DV	1.0
RA TO RA	2.0



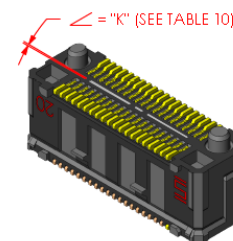
FINAL X AXIS ANGULAR MISALIGNMENT

TABLE 9			
POSITION	"J" = DEG	POSITION	"J" = DEG
5	4.0	30	0.7
10	2.0	40	0.6
20	1.0	50	0.5



FINAL Y AXIS ANGULAR MISALIGNMENT

TABLE 10			
POSITION	"K" = DEG	POSITION	"K" = DEG
5	1.4	30	0.2
10	0.6	40	0.1
20	0.3	50	0.1



FINAL Z AXIS ANGULAR MISALIGNMENT

7.3 Multiple Connector Applications: Due to the interlocking features and fine pitch of the LSXX (LSEM, LSS or LSHM) connectors, maintaining proper linear and angular alignment during mating and unmating is essential to minimize the potential for connector damage. As a result, Samtec highly recommends against the use of multiple LSXX connectors, mated simultaneously, per board. However, for applications requiring two or more simultaneously mated connectors per board, please review the [Multiple LSXX Connector Applications](#) document. For more information, please contact IPG@Samtec.com.

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7.4 Due to variances in equipment, solder pastes and applications (board design, component density, etc.), Samtec does not specify a recommended reflow profile for our connectors. The processing parameters provided by the solder paste manufacturer should be employed and can usually be found on their website.

All of Samtec's surface mount components are lead free reflow compatible and compliant with the profile parameters detailed in IPC/JEDEC J-STD-020 which requires that components be capable of withstanding a peak temperature of 260°C as well as 30 seconds above 255°C.

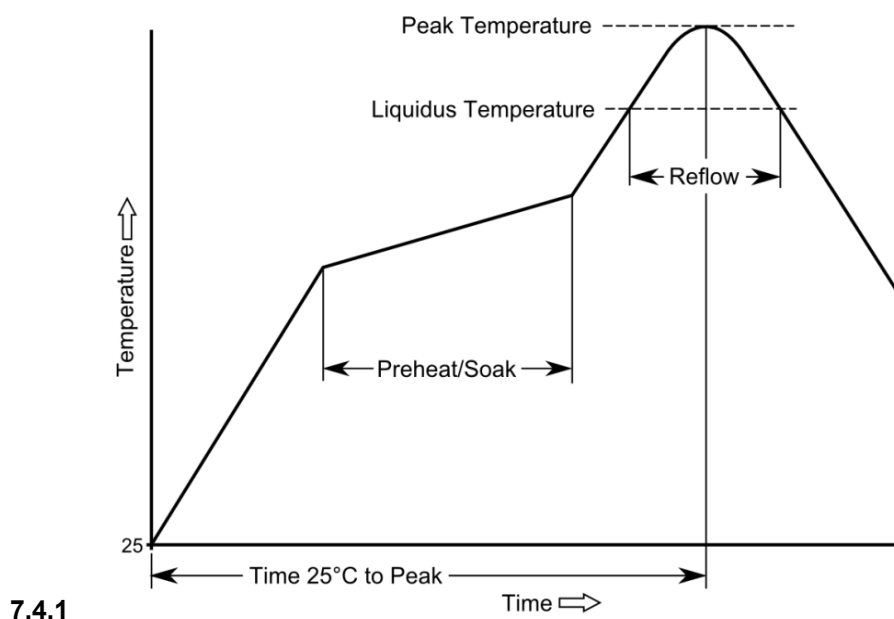
Samtec Recommended Temperature Profile Ranges (SMT)

Sn-Pb Eutectic Assembly

Preheat/Soak (100°C-150°C)	Max Ramp Up Rate	Reflow Time (above 183°C)	Peak Temp	Time within 5°C of 235°C	Max Ramp Down Rate	Time 25°C to Peak Temp
60-120 sec.	3°C/s max.	40-150 sec.	235°C	20 sec. max.	6°C/s max.	6 min. max.

Pb-Free Assembly

Preheat/Soak (150°C-200°C)	Max Ramp Up Rate	Reflow Time (above 217°C)	Peak Temp	Time within 5°C of 260°C	Max Ramp Down Rate	Time 25°C to Peak Temp
60-120 sec.	3°C/s max.	40-150 sec.	260°C	30 sec. max.	6°C/s max.	8 min. max.



These guidelines should not be considered design requirements for all applications.
Samtec recommends testing interconnects on your boards in your process to guarantee optimum results.

Series: LSHM 0,50mm (.0197") Razor Beam™ High Speed Hermaphroditic Terminal/Socket Strip

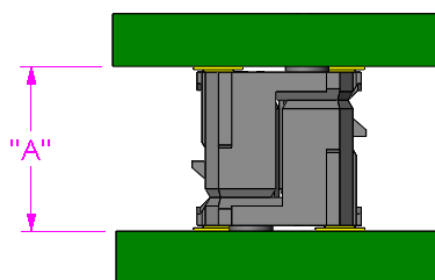
7.5 Maximum Reflow Passes: The parts can withstand three reflow passes at a maximum component temperature of 260°C.

7.6 Stencil Thickness: The recommended stencil thickness is .005" (0,13 mm).

7.7 Placement: Machine placement of the parts is strongly recommended.

7.8 Reflow Environment: Samtec recommends the use of a low level oxygen environment (typically achieved through Nitrogen gas infusion) in the reflow process to improve solderability.

7.9 Hardware: Board-to-board standoffs are recommended to provide a robust mechanical connection. Samtec offers several different types, which can be found here: [Samtec Standoff Offerings](#).



LSHM LEAD STYLE	LSHM LEAD STYLE							
	-02.5		-03.0		-04.0		-06.0	
	Nominal Connector Stack Height "A" in [mm]	Standoff	Nominal Connector Stack Height "A" in [mm]	Standoff	Nominal Connector Stack Height "A" in [mm]	Standoff	Nominal Connector Stack Height "A" in [mm]	Standoff
-02.5	.197 [5.00]	XXXX-0515	N/A	N/A	N/A	N/A	N/A	N/A
-03.0	.217 [5.50]	XXXX-0565	.236 [6.00]	XXXX-0615	N/A	N/A	N/A	N/A
-04.0	.256 [6.50]	XXXX-0665	.276 [7.00]	XXXX-0715	.315 [8.00]	XXXX-0815	N/A	N/A
-06.0	.335 [8.50]	XXXX-0865	.354 [9.00]	XXXX-0915	.394 [10.00]	XXXX-1015	.472 [12.00]	XXXX-1215

XXXX-XXXX

BOARD STACK HEIGHT

STANDOFF SERIES

Series: LSHM 0,50mm (.0197") Razor Beam™ High Speed Hermaphroditic Terminal/Socket Strip

- 7.10 Cleaning:** Samtec, Inc. has verified that our connectors may be cleaned in accordance with the solvents and conditions designated in the EIA-364-11 standard.
- 7.11 Inverted Reflow:** For applications of right angle (-DH) or reverse right angle (-RH) configurations requiring inverted reflow soldering, use of an epoxy may be necessary. For more information on the use of epoxy adhesives, please visit the Processing page on our website or contact Samtec's Interconnect Processing Group at IPG@samtec.com

8.0 ADDITIONAL RESOURCES

- 8.1** For additional mechanical testing or product information, contact our Customer Engineering Support Group at CES@samtec.com
- 8.2** For additional information on high speed performance testing, contact our Signal Integrity Group at SIG@samtec.com
- 8.3** For additional processing information, contact our Interconnect Processing Group at IPG@samtec.com
- 8.4** For RoHS, REACH or other environmental compliance information, contact our Product Environmental Compliance Group at PEC@samtec.com

USE OF PRODUCT SPECIFICATION SHEET

This Product Specification Sheet ("PSS") is a brief summary of information related to the Product identified. As a summary, it should only be used for the limited purpose of considering the purchase/use of Product. For specific, detailed information, including but not limited to testing and Product footprint, refer to Section 2.0 of this document and the links there provided to test reports and prints. This PSS is the property of Samtec, Inc. ("Samtec") and contains proprietary information of Samtec, our various licensors, or both. Samtec does not grant express or implied rights or license under any patent, copyright, trademark or other proprietary rights and the use of the PSS for building, reverse engineering or replication is strictly prohibited. By using the PSS, the user agrees to not infringe, directly or indirectly, upon any intellectual property rights of Samtec and acknowledges that Samtec, our various licensors, or both own all intellectual property therein. The PSS is presented "AS IS". While Samtec makes every effort to present excellent information, the PSS is only provided as a guideline and does not, therefore, warrant it is without error or defect or that the PSS contains all necessary and/or relevant information about the Product. The user agrees that all access and use of the PSS is at its own risk. **NO WARRANTIES EXPRESSED OR IMPLIED, INCLUDING ANY WARRANTY OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE OR OF ANY KIND WHATSOEVER ARE PROVIDED.**